Jayadithya R

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GENAI ENGINEER, LLM FINE-TUNING, TOKENIZER INFRA, AND EVAL

Builds AI systems end to end: data, training, inference, evaluation. MLIR compiler work at Blubridge Technologies, a Rust BPE tokenizer published on Hugging Face, a first-author IEEE paper on vision-language distillation, and million-row data pipelines.

SKILLS

Languages: Python | TypeScript | SQL | Rust | Bash

AI/ML: LoRA fine-tuning (Unsloth) | Hugging Face Hub and Datasets | open-model inference (Gemma, Qwen) | CLIP / transformers.js | ChromaDB | scikit-learn

Backend and Data: FastAPI | PostgreSQL | SQLite | Meilisearch | Docker | Docker Compose | GitHub Actions | scrapy | yt-dlp

Spoken Languages: Tamil (native) | English (proficient) | Hindi (basic) | Telugu (basic)

EDUCATION

B.Tech Artificial Intelligence and Data Science, Dr. M.G.R. Educational and Research Institute, Chennai, TN | 2026

PROJECTS

tamil-tokenizer | *Rust, rayon, Python, Hugging Face* (github.com/barryallen16/tamil-tokenizer)
- Designed a 1,135-token Tamil-aware base vocabulary in Rust around conjunct stems and ligatures, after measuring that conjuncts, not characters, drive Tamil compression
- Published Tamil BPE models that beat mBERT and Llama-3 on held-out compression; re-indexed the merge table to keep the larger vocabulary fast

fitcheck-family | *Qwen3-VL, LoRA, FashionCLIP, Next.js, Postgres* (github.com/barryallen16/fitcheck-website)
- Ran the teacher-student distillation behind a first-author IEEE paper: a Qwen3-VL-8B teacher annotated 106K garment images, yielding 538K caption-and-QA training pairs
- Distilled the teacher into an on-device student: 3.1x less VRAM at 92% of teacher quality, validated against a blind human study

VulnChecker-Java | *Python, Unsloth, LoRA, Qwen2.5* (github.com/barryallen16/VulnChecker-Java)
- LoRA fine-tuned Qwen2.5-7B over a 7-stage corpus pipeline, reaching 97% on a held-out Java vulnerability set; datasets and eval harness published on Hugging Face

kaggle-automation | *Python, FastAPI, Docker, GitHub Actions, AWS EC2* (github.com/barryallen16/kaggle-automation)
- Dockerized a FastAPI control plane on Amazon Web Services (AWS) that pools 19 Kaggle credentials to shard GPU jobs across accounts, with quota meters and Telegram alerts

sivabharani-comments | *Python, Gemma, Hugging Face Datasets* (github.com/barryallen16/sivabharani-comments)
- Built a resumable pipeline over 2,755 Tamil tech videos, then ran local Gemma 4-bit inference across 1.18M transcript chunks

WORK EXPERIENCE

Blubridge Technologies, Chennai, TN | AI Research Intern | 07/2025 - 10/2025
- Implemented operator-fusion passes in MLIR for the team's custom tensor dialect, cutting kernel launch overhead by roughly 20% on benchmark workloads
- Added lowering patterns from the high-level dialect down to LLVM IR, including shape inference and bufferization for dynamic tensors
- Wrote 40+ lit tests covering pass correctness and IR round-trip stability, wired into the team's CI